

Fitness Tracker Data Analysis Report

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1. Data Setup & Summary Statistics

This section prepares the R environment, loads the dataset, converts dates, and computes descriptive statistics for activity and sleep. The dataset includes daily activity and sleep logs for multiple users.

1.1. R Setup and Data Loading

```
# Install and load tidyverse
install.packages("tidyverse")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.2      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.1

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

# Load the dataset
data <- read_csv("final_analysis_set.csv")

## Rows: 622 Columns: 13
## -- Column specification -----
## Delimiter: ","
## db1 (12): Id, TotalSteps, TotalDistance, VeryActiveMinutes, FairlyActiveMin...
## date (1): ActivityDate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

# Convert ActivityDate to Date format
data <- data %>%
  mutate(ActivityDate = as.Date(ActivityDate))
```

1.2. Summary Statistics (Activity + Sleep + Dataset Structure)

```
summary_stats <- data %>%
  summarise(
    Unique_Users = n_distinct(Id),
    Unique_Dates = n_distinct(ActivityDate),

    Avg_Steps = mean(TotalSteps, na.rm = TRUE),
    Median_Steps = median(TotalSteps, na.rm = TRUE),
    Max_Steps = max(TotalSteps, na.rm = TRUE),

    Avg_Calories = mean(Calories, na.rm = TRUE),
    Median_Calories = median(Calories, na.rm = TRUE),
    Max_Calories = max(Calories, na.rm = TRUE),

    Avg_Sleep_Minutes = mean(minutes_asleep, na.rm = TRUE),
    Median_Sleep_Minutes = median(minutes_asleep, na.rm = TRUE),
    Avg_Sleep_Efficiency = mean(sleep_efficiency_pct, na.rm = TRUE)
  )

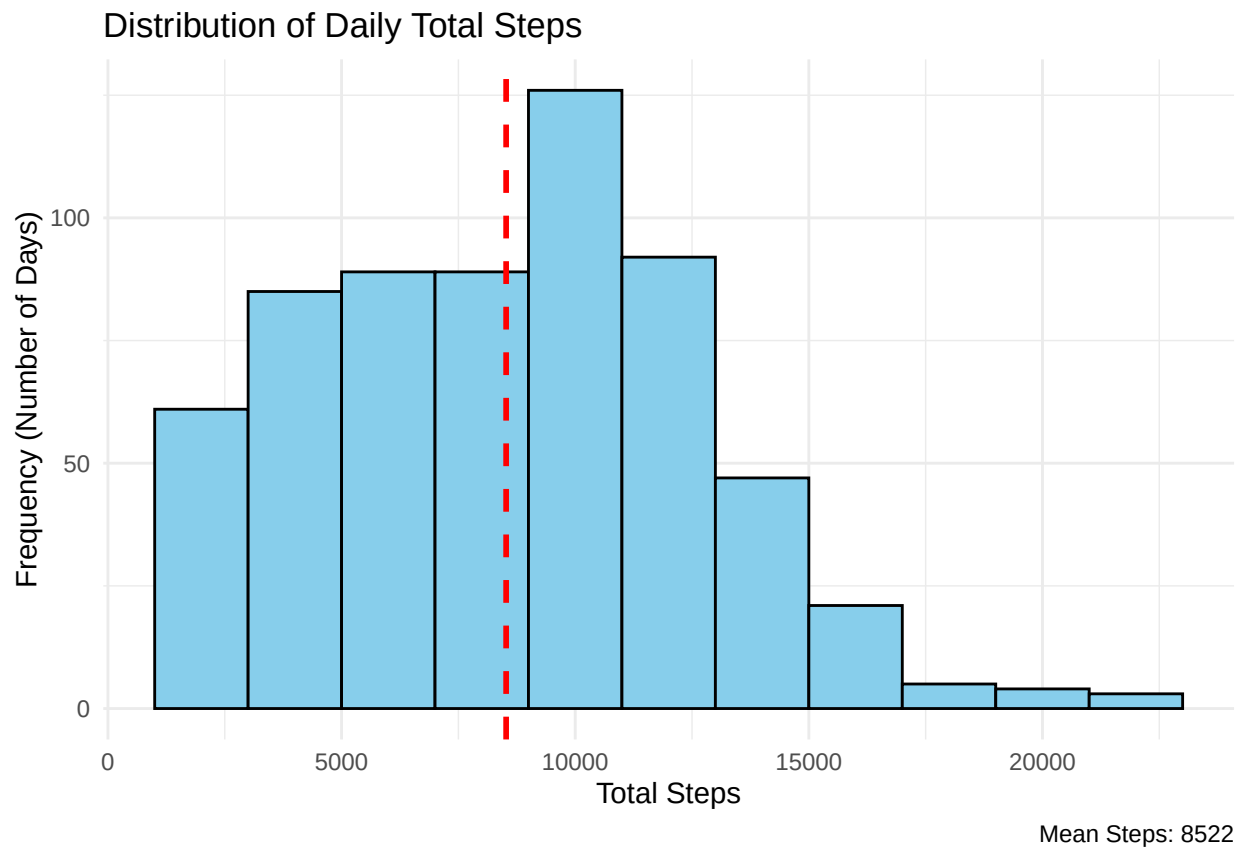
print(summary_stats, width = Inf)

## # A tibble: 1 x 11
##   Unique_Users Unique_Dates Avg_Steps Median_Steps Max_Steps Avg_Calories
##   <int>         <int>      <dbl>      <dbl>      <dbl>      <dbl>
## 1          24         55    8522.      8776.     22770     2399.
##   Median_Calories Max_Calories Avg_Sleep_Minutes Median_Sleep_Minutes
##   <dbl>         <dbl>      <dbl>      <dbl>
## 1          2228         4900        397.        415
##   Avg_Sleep_Efficiency
##   <dbl>
## 1          91.2
```

2. Visualizations and Insights

2.1. Distribution of Daily Total Steps

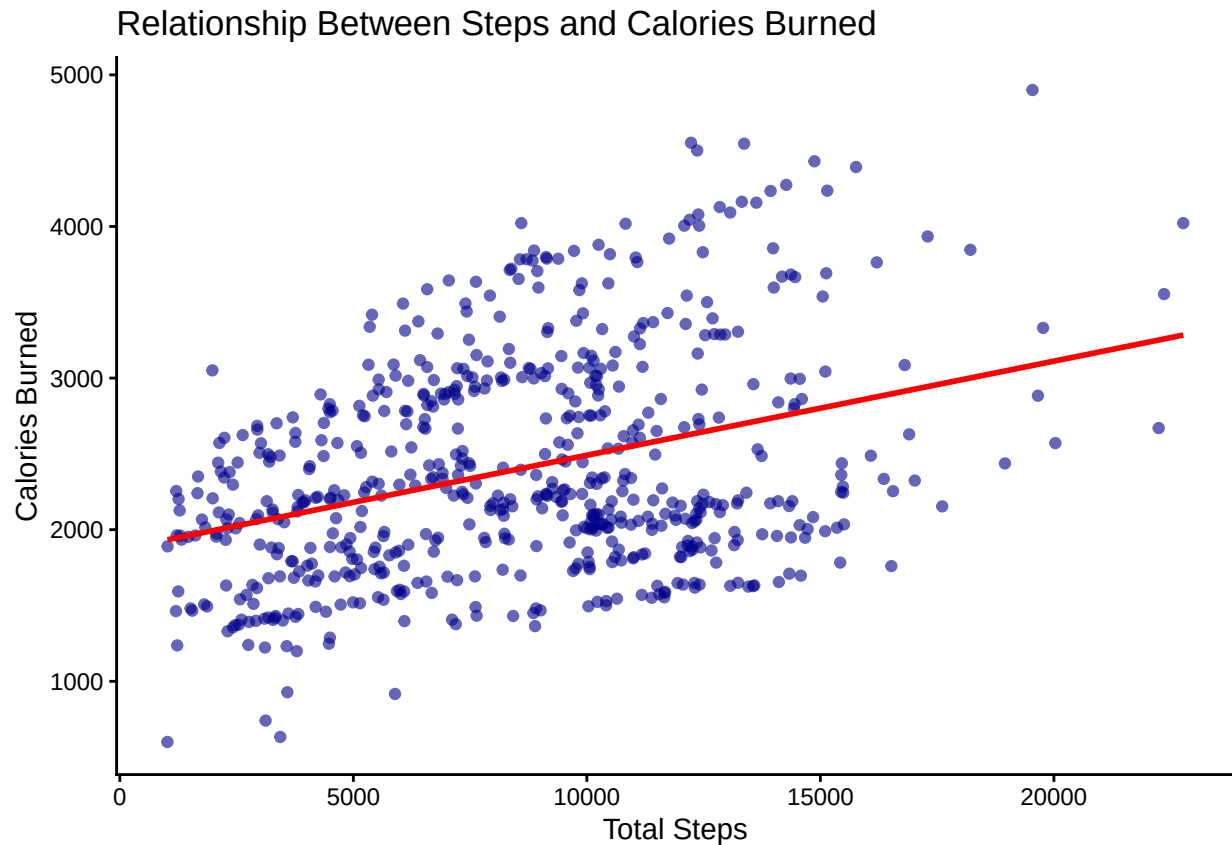
```
ggplot(data, aes(x = TotalSteps)) +
  geom_histogram(binwidth = 2000, fill = "skyblue", color = "black") +
  geom_vline(aes(xintercept = mean(TotalSteps, na.rm = TRUE)),
    linetype = "dashed", color = "red", linewidth = 1) +
  labs(
    title = "Distribution of Daily Total Steps",
    x = "Total Steps",
    y = "Frequency (Number of Days)",
    caption = paste("Mean Steps:", round(mean(data$TotalSteps), 0))
  ) +
  theme_minimal()
```



2.2. Relationship Between Steps and Calories Burned

```
ggplot(data, aes(x = TotalSteps, y = Calories)) +  
  geom_point(alpha = 0.6, color = "darkblue") +  
  geom_smooth(method = "lm", color = "red", se = FALSE) +  
  labs(  
    title = "Relationship Between Steps and Calories Burned",  
    x = "Total Steps",  
    y = "Calories Burned"  
  ) +  
  theme_classic()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



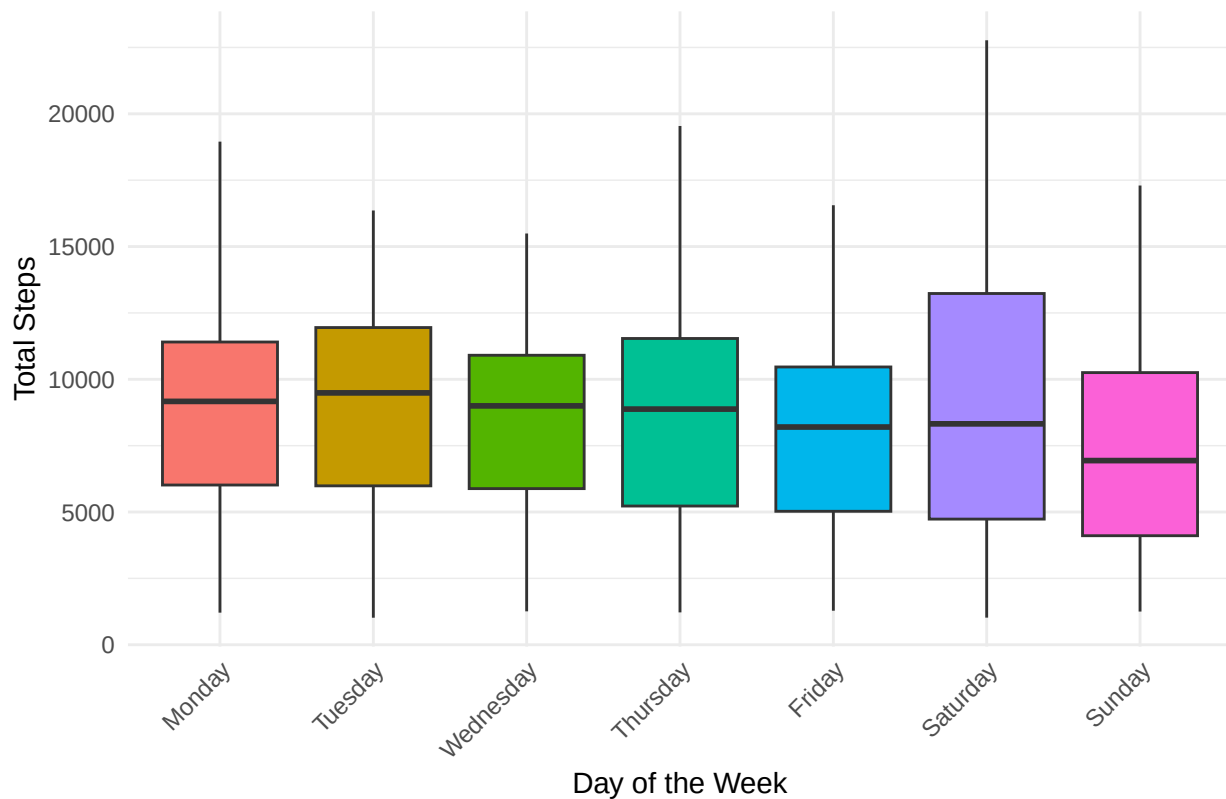
3. Weekly Trends

3.1. Boxplot: Total Steps by Day of the Week

```
data <- data %>%
  mutate(
    WeekDay = weekdays(ActivityDate),
    WeekDay = factor(WeekDay,
                     levels = c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"))
  )

ggplot(data, aes(x = WeekDay, y = TotalSteps, fill = WeekDay)) +
  geom_boxplot() +
  labs(
    title = "Distribution of Total Steps by Day of the Week",
    x = "Day of the Week",
    y = "Total Steps"
  ) +
  theme_minimal() +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1))
```

Distribution of Total Steps by Day of the Week



3.2. Average Steps by Day

```
daily_step_average <- data %>%
  group_by(WeekDay) %>%
  summarise(
    Avg_Steps = round(mean(TotalSteps, na.rm = TRUE), 0),
    Median_Steps = round(median(TotalSteps, na.rm = TRUE), 0),
    Total_Observations = n()
  )

print(daily_step_average)
```

```
## # A tibble: 7 x 4
##   WeekDay Avg_Steps Median_Steps Total_Observations
##   <fct>    <dbl>      <dbl>          <int>
## 1 Monday      8662        9167             81
## 2 Tuesday      8971        9487             95
## 3 Wednesday    8382        8998             90
## 4 Thursday    8558        8877             90
## 5 Friday       8005        8202             86
## 6 Saturday    9396        8321             94
## 7 Sunday       7562        6937             86
```

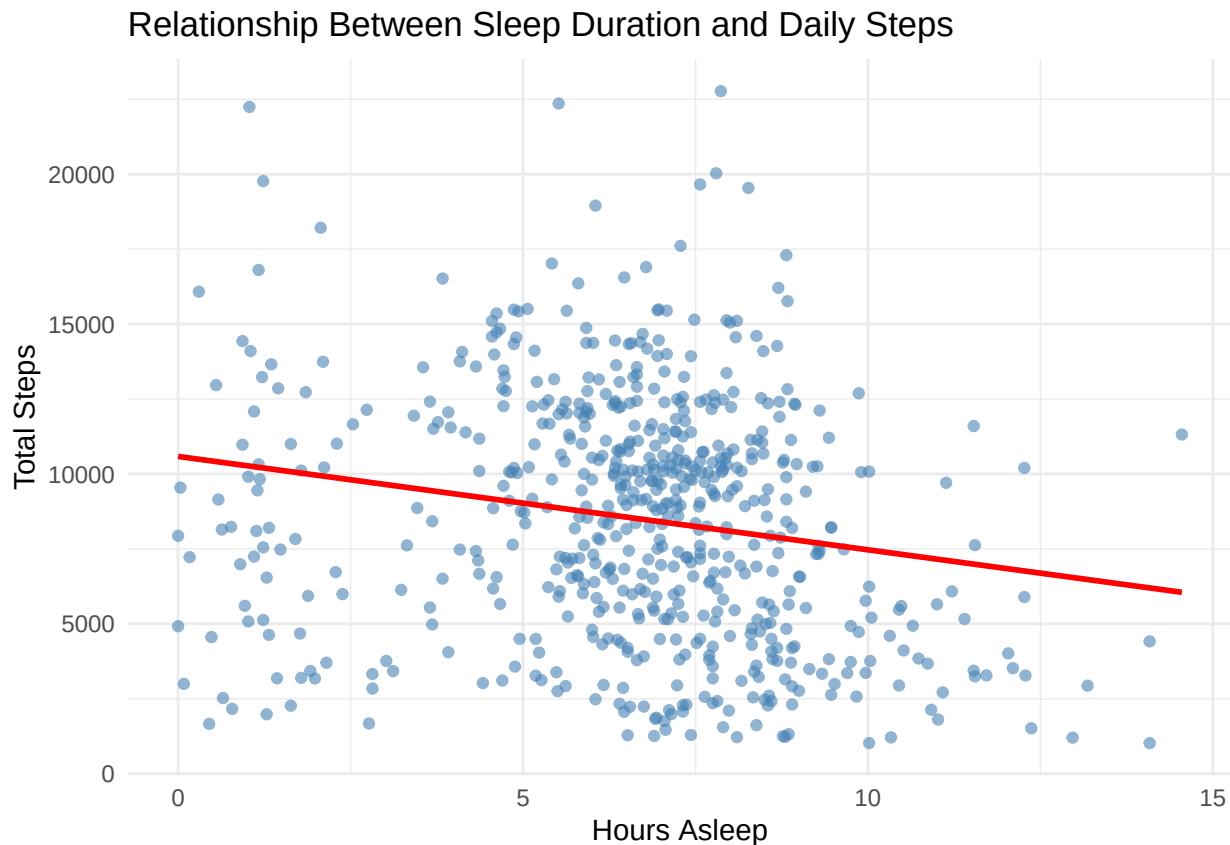
4. Sleep Analysis

4.1. Steps vs Hours Asleep

```
data$hours_asleep <- data$minutes_asleep / 60

ggplot(data, aes(x = hours_asleep, y = TotalSteps)) +
  geom_point(color = "steelblue", alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(
    title = "Relationship Between Sleep Duration and Daily Steps",
    x = "Hours Asleep",
    y = "Total Steps"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



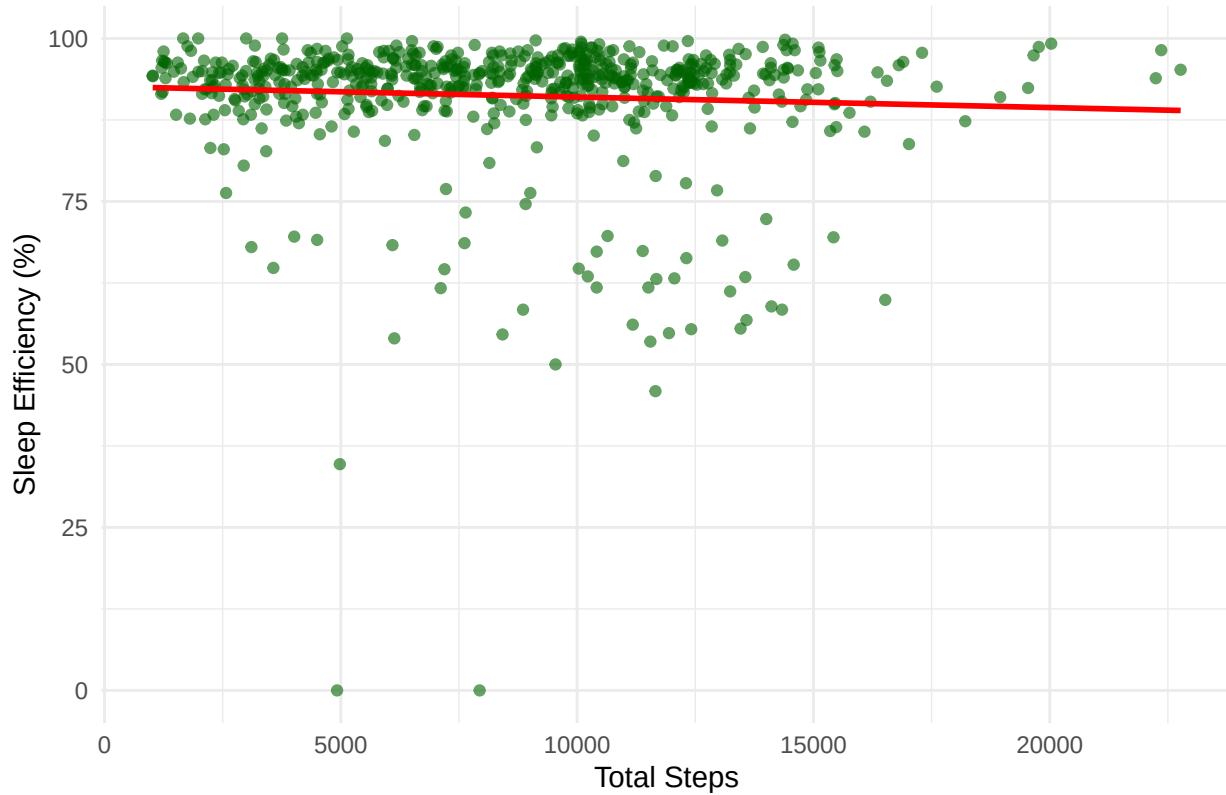
4.2. Steps vs Sleep Efficiency

```
ggplot(data, aes(x = TotalSteps, y = sleep_efficiency_pct)) +
  geom_point(color = "darkgreen", alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(
    title = "Relationship Between Steps and Sleep Efficiency",
    x = "Total Steps",
    y = "Sleep Efficiency (%)"
  )
```

```
) +  
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Relationship Between Steps and Sleep Efficiency



```
library(reshape2)
```

```
##  
## Attaching package: 'reshape2'  
## The following object is masked from 'package:tidyr':  
##  
## smiths
```

```
corr_data <- data %>%  
  select(Steps = TotalSteps,  
         Cal = Calories,  
         SleepMin = minutes_asleep,  
         BedTime = total_time_in_bed,  
         Eff = sleep_efficiency_pct)  
  
corr_matrix <- cor(corr_data, use = "complete.obs")  
  
ggplot(melt(corr_matrix), aes(Var1, Var2, fill = value)) +  
  geom_tile() +  
  scale_fill_gradient2(low = "red", high = "blue", mid = "white", midpoint = 0) +  
  labs(title = "Correlation Matrix: Activity & Sleep Metrics") +  
  theme_minimal()
```

